Dynamic Programming

Part 3: Exercise 5 (week 17)

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Plan for today

- Discrete-Continuous choice models
- Consumption-Saving + Discrete Retirement choice model
- DC-EGM algortihm

Consumption-Saving + Discrete Retirement Choice model

Deaton's model + Retirement choice $\rightarrow$ New Bellman Equation:

$$v_t(m_t, z_t, \epsilon_t(0), \epsilon_t(1)) = \max_{z_{t+1} \in \mathcal{Z}(z_t)} \{\mathcal{V}_t(m_t, z_{t+1}) + \sigma_\epsilon \epsilon_t(z_{t+1})\}$$

$$\text{s.t. } \mathcal{Z}(z_t) = \begin{cases} \{0, 1\} & z_t = 0 \\ \{1\} & z_t = 1 \end{cases}$$

$$\epsilon_t(0), \epsilon_t(1) \sim Gumbel()$$

Caution: z_t is "retirement state" and z_{t+1} is "choice" in period t of retirement state in period $t+1$

$\epsilon_t(z_{t+1})$ is taste-shock associated with "choice" z_{t+1}

Direction in z_t state: Once retired $z_t = 1 \rightarrow$ Can't "unretire"

Retirement model: Choice-specific value functions

Sub-problem for each discrete choice mimicks known structure

$$\mathcal{V}_t(m_t, z_{t+1}) = \max_{C_t} \left\{ \frac{C_t^{1-\rho}}{1-\rho} - \alpha 1(z_{t+1} = 0) + \beta \mathbb{E}_t[v_{t+1}(m_{t+1}, z_{t+1}, \epsilon_{t+1}(0), \epsilon_{t+1}(1))] \right\}$$

$$\text{s.t. } m_{t+1} = R(m_t - c_t) + W\xi_{t+1} 1(z_{t+1} = 0)$$

$$a_t = m_t - c_t \geq 0$$

$$\xi_{t+1} \sim \exp(\mathcal{N}(-0.5\sigma_\xi^2, \sigma_\xi^2))$$

New stuff: disutility of working, only recieve wage when working

Value function had max over discrete choice $z_{t+1} \in \mathcal{Z}(z_t)$

Choice-specific value function has max over continuous choice C_t

$\rightarrow$ Dif' under max $\rightarrow$ **Non-sufficient but necessary Euler Equation:**

$$C_t^{*-\rho} = \beta R \mathbb{E}_t[C_{t+1}^{*-\rho}]$$

DC-EGM algorithm

Idea: Use all (cond'l) Euler Equations and choose upper envelope

Period T: Solve (trivial) problem: $c_T^*(m_T, z_T, \dots) = m_T$

Period $t = T-1, T-2, \dots, 1$:

Not-retired in t: $z_t = 0$. Loop over discrete choices z_{t+1}

Compute $\mathbb{E}_t[c_{t+1}^{*-\rho} | z_{t+1}]$

Compute (cond'l) policies: $\tilde{c}_t = \left(\beta R \mathbb{E}_t[c_{t+1}^{*-\rho} | z_{t+1}]\right)^{-\frac{1}{\rho}}$

(and implied cash-on-hand $\tilde{m}_t$)

<u>Problem: Euler Equation is ONLY necessary!</u>

Implication: $\tilde{c}_t$ can be sub-optimal $\rightarrow$ Sort + clean (next slide)

Retired in t: $z_t = 1$ $\rightarrow$ no discrete choice (can't unretire)

$\rightarrow$ Standard EGM: Euler Equation is sufficient again!

DC-EGM: Upper envelope

$\tilde{c}_t$ can be sub-optimal? Zig-zag, non-uniqueness in Euler Equation

From EGM-step, we have raw points $(\tilde{m}_t, \tilde{c}_t, \tilde{\mathcal{V}}_t)$

- Loop over raw endo. grid (potentially non-ascending): $\tilde{m}_t^i$
  - Compute $c_{slope} = \frac{\tilde{c}_t^{i+1} - \tilde{c}_t^i}{\tilde{m}_t^{i+1} - \tilde{m}_t^i}$
  - Loop over sorted endo. grid: $\tilde{m}_t^{s,j}$
    - Compute $c_t^{guess,j} = \tilde{c}_t^i + c_{slope}(\tilde{m}_t^{s,j} - \tilde{m}_t^i)$
    - Compute choice-specific value given $c_t^{guess,j}$
    - Replace choice-specific value (and policy) if bigger

$\rightarrow$ Way of removing lower zig-zags! $\rightarrow$ Upper envelope!

Your time to shine!